

BAIXIN (MAX) GUO

Human-Centered AI | Feedback Learning | Reliable AI Agents

gbx1220max@gmail.com | Website | GitHub

China | Remote availability from October 2026

RESEARCH PROFILE

I am an Applied Psychology graduate working on human-centered AI, feedback modeling, and agent evaluation. I design evaluation methods, inspect what feedback actually measures, and build reproducible research tools. My current work connects behavioral signals and individual preferences with the reliability of AI decisions.

Interests: Human and evaluator feedback; preference learning; human-agent collaboration; reliable adaptation; reproducible AI evaluation.

RESEARCH COLLABORATIONS

Implicit human feedback | Sep 2026 - present

Research collaboration with Haw-Shiuan Chang, University of Arizona

Ongoing collaboration | data and representation audit

- Audited missing and duplicate events, coordinate validity, temporal gaps, and query-trajectory-response alignment.
- Compared word-run and fixation-proxy representations and examined how preprocessing choices change the signal. Learning improvements have not yet been established.

Memory-agent evaluation reliability | Sep 2026

Research collaboration with Dino Vitale, Boundary Labs

Delivered evaluation framework | bounded trace review

- Delivered an observability map, failure-attribution framework, five diagnostic tests, and a judge-reliability protocol.
- Distinguished independently inspected traces from collaborator-reported checks; limited attribution claims to the evidence actually available.

SELECTED RESEARCH & TECHNICAL WORK

Preference Modeling Diagnostics | 2026

Released | frozen diagnostic study

- Designed prompt-disjoint evaluation and identity controls to distinguish stable individual preference signal from leakage.
- Regularized adaptation reduced equal-user negative log-likelihood by 0.879% across 94 eligible known users. Kept the confirmatory test separate from exploratory history-length analysis.

[Public artifact](#)

Evaluation Runtime | 2026

Public research prototype | deterministic mock agent

- Specified and co-designed attempt identity, evaluator-version binding, ambiguity-aware recovery, and denominator-aware reports.
- Directed implementation and reviewed reproducibility checks. The public snapshot uses deterministic mocks; software checks do not establish real-provider reliability.

[Public artifact](#)

SELECTED RESEARCH & TECHNICAL WORK - CONTINUED

CheckMyCoach | 2026

Prototype | human validation in preparation

- Co-designed the evaluation decomposition, specified routing and revision behavior, and reviewed outputs against explicit preservation requirements.
- In a fixed 40-case constructed development run, 15 cases were routed and 12 delivered outputs removed the target feature; none passed every target-and-retention check. No human-outcome improvement is claimed.

Public artifact

OptiFrontier | 2026

Bounded optimizer-architecture experiment

- Ran a prespecified 6,000-step optimizer comparison on a 162M-parameter GPT-style model with a fixed architecture on an A100 GPU.
- Documented a trainability gap within that one-seed configuration and packaged commands, logs, hashes, and a bounded decision record.

SkillHone contributions | 2026

Open-source contributions | three PRs awaiting review

- Submitted regression examples with reproducible failures, repair workflows, and verification checks.
- PRs #14, #15, and #16 are open and awaiting review as of October 1, 2026.

PRs #14, #15, #16

Undergraduate thesis - Chinese LLM evaluation on CBT-Bench | 2026

Benchmark evaluation

- Designed fixed zero-shot evaluation across five Chinese LLM services on 442 structured examples, with checkpointing and retry logic.
- Reported exact match, partial-overlap, and sample-average F1; bounded findings to benchmark evidence rather than clinical or human-outcome validity.

EARLIER RESEARCH EXPERIENCE

Trojan Sports Research Lab, USC Marshall School of Business | Mar 2025 - Mar 2026

Research collaboration | Supervisor: Prof. Lorena Martin

- Collected, cleaned, visualized, and analyzed multi-source baseball data for research on pitcher workload, pitch characteristics, and injury risk.

EDUCATION

Changchun Humanities and Sciences College | 2022-2026

Bachelor's degree in Applied Psychology

METHODS & TECHNICAL SKILLS

Programming: Python; JavaScript/TypeScript; SQL/SQLite; Git/GitHub; pytest; CLI tooling; structured data pipelines.

AI and evaluation: LLM/API pipelines; agent prototyping; supervised ML; preference diagnostics; reproducible evaluation; contextual bandits.

Research: experimental design; psychometrics; statistical analysis; inter-rater reliability; behavioral-data quality and representation audits.